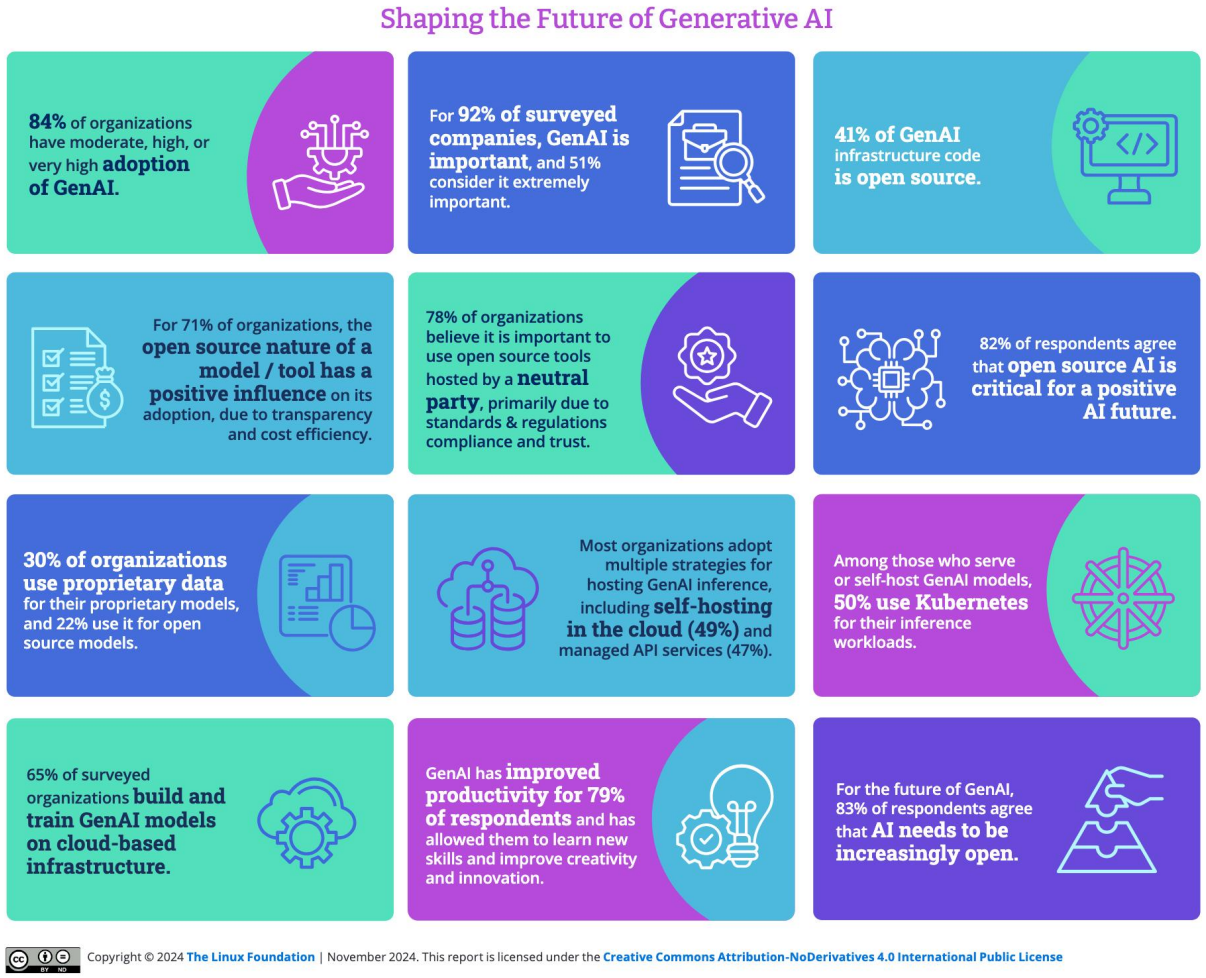
云原生+AI基础设施
Cloud Native AI Infra

开源创新的影响：塑造生成式人工智能的未来

我们知道 GenAI 正在以前所未有的速度改变行业。随着这项技术进入主流， 各组织都一致认为 AI 的未来必须是开放的。事实上，82% 的组织认为开源 AI 对于确保 AI 的积极未来至关重要，83% 的组织同意 AI 需要越来越开放，以促进信任、协作和 创新。

https://www.linuxfoundation.org/hubfs/LF%20Research/lfr_genai24_111924.pdf?hsLang=en



云原生AI Infra: 人工智能和云原生助力企业 IT 的未来

云原生人工智能是云原生不断发展的扩展。 云原生人工智能 (CNAI) 是指使用云原生原则构建和部署 AI 应用程序和工作负载的方法和模式。启用可重复和可扩展的以 AI 为中心的工作流程。

Cloud Native Artificial Intelligence is an evolving extension of Cloud Native.

Cloud Native Artificial Intelligence (CNAI) refers to approaches and patterns for building and deploying AI applications and workloads using the principles of Cloud Native. Enabling repeatable and scalable AI-focused workflows allows AI practitioners to focus on their domain.

以下是预测性人工智能和生成式人工智能在计算、网络和存储方面具有不同需求的一系列示例：

Below is a sample set of areas where predictive and generative AI have distinct needs across computing, networking, and storage:

Challenges/Need	Generative AI	Predictive AI
Computational Power	Extremely high. Requires specialized hardware.	Moderate to high. General-purpose hardware can suffice.
Data Volume and Diversity	Massive, diverse datasets for training.	Specific historical data for prediction.
Model Training and Fine-tuning	Complex, iterative training with specialized compute.	Moderate training.
Scalability and Elasticity	Highly scalable and elastic infrastructure (variable and intensive computational demands)	Scalability is necessary but lower elasticity demands. Batch processing or event-driven tasks.
Storage and Throughput	High-performance storage with excellent throughput. Diverse data types. Requires high throughput and low-latency access to data.	Efficient storage with moderate throughput. It focuses more on data analysis and less on data generation; data is mostly structured.
Networking	High bandwidth and low latency for data transfer and model synchronization (e.g., during distributed training).	Consistent and reliable connectivity for data access.

STATE OF CLOUD NATIVE DEVELOPMENT Q3 2025

/DATA

CLOUD NATIVE COMPUTING FOUNDATION

OCTOBER 2025

15.6M

CLOUD NATIVE DEVELOPERS IN Q3 2025

7.1M

ML/AI DEVELOPERS ARE CLOUD NATIVE

30%

OF ALL DEVELOPERS ARE DEPLOYING TO HYBRID CLOUD IN Q3 2025, SHOWING A CONTINUED INCREASE IN POPULARITY

Kubernetes在人工智能工作负载中日益普及的应用凸显了通用标准的重要性。

82%的组织已在构建定制化人工智能解决方案，其中58%采用Kubernetes来支持这些工作负载。

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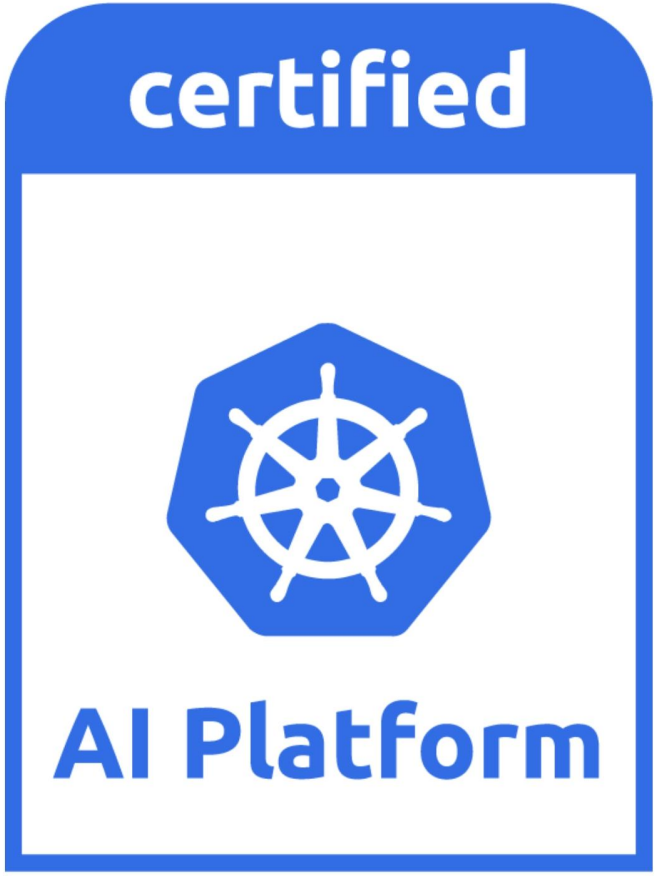
“随着AI在生产领域的持续扩展，并利用多云及系统资源，团队需要可靠的基础设施作为支撑，”CNCF CTO Chris 表示。

“该合规计划将建立统一标准，确保AI工作负载在不同环境中表现可预测。它基于我们与Kubernetes成功采用的社区驱动流程，助力在AI应用规模扩展时，实现100多个Kubernetes系统的统一性。”

https://github.com/cncf/k8s-ai-conformance

Certified AI Platform Conformance Program



这一新举措使CNCF将其经过验证的模型应用于人工智能基础设施。目标是通过使用开放、标准的API和接口，为运行遵循Kubernetes原则的AI任务设定明确的要求，从而减少混乱和不一致。通过制定明确的标准、测试合规性并在社区内达成协议，CNCF旨在加快人工智能的使用，同时降低风险。这种方法反映了Kubernetes所采用的成功策略，现在已经适应了快速发展的人工智能领域。

主要目标

- 简化Kubernetes上的AI/ML并加速采用
- 保证AI工作负载的互操作性和可移植性
- 在行业标准的基础上实现人工智能工具的生态系统增长

*Image generated by Gemini

100多个 Conformance Test

- Accelerators
- Storage
- Networking
- Scheduling and Orchestration
- Observability and Telemetry
- Security
- Maintenancem & Repairs
- AI Framwork & Operator Enablement

Requirements

- **MUST** indicates a mandatory requirement for conformance.
- **SHOULD** indicates a recommended practice or a desired aspirational goal. While not mandatory, it signifies a course of action that the community is moving towards and believes will lead to better outcomes, improved interoperability, or enhanced functionality. It is anticipated that SHOULD requirements may evolve into MUST requirements in future iterations of a standard as practices mature and consensus solidifies around their universal benefit.

The platform should support enabling all required components, even those not part of standard Kubernetes, eliminating the need for users to manually install these components separately.

Accelerators	**Accelerator Resource Exposure & Allocation**	**MUST**: Support Dynamic Resource Allocation (DRA) APIs to enable more flexible and fine-grained resource requests beyond simple counts. **How we might test it:** Verify that all the resource.k8s.io/v1 DRA API resources are enabled.
	Accelerator Driver & Runtime Management	**SHOULD**: Provide a verifiable mechanism for ensuring that compatible accelerator drivers and corresponding container runtime configurations are correctly installed and maintained on nodes with accelerators.
		SHOULD: Advertise the versions of the installed accelerator drivers, which could impact workload compatibility. This version information must be exposed as DRA attributes (e.g. `dra.vendor.com/driver-version`)
	GPU Sharing	**SHOULD**: If accelerator supports GPU sharing, provide well-defined mechanisms for GPU sharing to improve utilization for workloads that do not require a full dedicated GPU. If hardware-level partitioning is supported, these fractional GPU resources should be exposed as distinct schedulable resources via DRA once it's supported. If software-based sharing (e.g. time-slicing) is supported, device plugin should be used to allow oversubscription of GPUs, with a forward-looking path to DRA.
	Virtualized Accelerators	**SHOULD**: If the platform supports virtualized accelerator technologies (e.g. vGPU), these should also be exposable and manageable via DRA (once it's supported), maintaining consistency with physical fractional GPUs.
	Hardware Topology Awareness	**SHOULD**: For nodes with specialized hardware topologies, information about the physical layout of accelerators and their proximity to high-performance network interfaces (NICs) should be discoverable in a structured manner. This includes, but is not limited to: ● Direct high-speed interconnects between accelerators ● The alignment of accelerators and NICs on the node's physical topology This information should be exposed via DRA attributes if supported by accelerator and driver to enable topology-aware scheduling.

PROJECTS

hami

HAMi

Heterogeneous AI Computing Virtualization Middleware

hami was accepted to CNCF on August 21, 2024 at the **Sandbox** maturity level.

VISIT PROJECT WEBSITE

hami case studies

RELATED CASE STUDIES

SF Technology

Effective GPU: A Heterogeneous AI Virtualization Pooling solution by SF technology built on top of HAMi

Up to 57%

GPU Savings for production and test clusters

September 18, 2025

READ CASE STUDY

Prep EDU

Enabling Efficient GPU Orchestration for AI Inference at PREP EDU

90%

of GPU infra optimized using HAMi

August 20, 2025

READ CASE STUDY

PROJECT INSIGHTS

Key metrics, providing insights into development activity, community engagement, and project health. Powered by **LFX Insights**.

HEALTH SCORE

Excellent (82)

Health Score measures a project's overall trustworthiness across four key areas: contributors, development, popularity, and security.

TOTAL CONTRIBUTORS

3,498

+166% vs. previous year

GITHUB STARS

3,232

+179% vs. previous year

SOFTWARE VALUE

$41.1M

TOTAL CONTRIBUTING ORGANIZATIONS

617

+147% vs. previous year

GITHUB FORKS

700

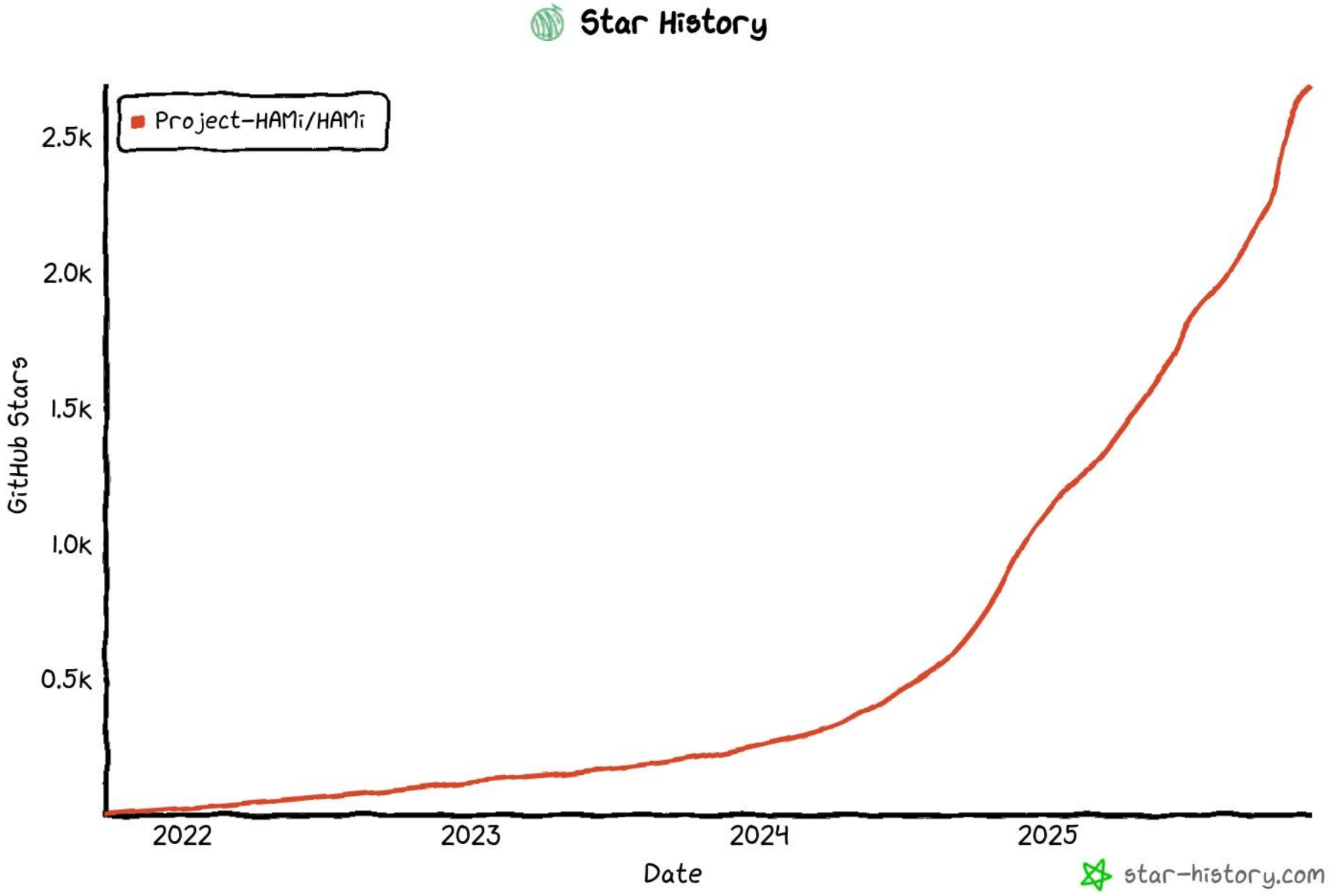
+176% vs. previous year

FIRST COMMIT

March 15, 2020

EXPLORE MORE INSIGHTS

Star History



谢谢